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Introduction

This proposal presents an analytical study of UIDAI Aadhaar enrolment, demographic update, and biometric update datasets to evaluate the operational health of India's national digital identity infrastructure. The objective of the project is to move beyond basic counts and statistics and extract meaningful, actionable insights related to identity maintenance, update behavior, and operational consistency across states and districts.

The analysis applies exploratory data analysis, engineered metrics, and selective machine learning techniques to assess how actively Aadhaar identities are maintained after enrolment. Rather than treating Aadhaar as a static registry, this study models it as a continuously evolving identity ecosystem.

Problem Statement

Although Aadhaar enrolment has achieved near-universal coverage, a significant volume of demographic and biometric updates continues to be observed across regions. These updates are influenced by factors such as internal migration, occupational mobility, population aging, and service accessibility.

A major challenge is the lack of consolidated analytical indicators that capture the *health* of identity records after enrolment. Simple enrolment counts alone do not reflect identity freshness, operational stress, or maintenance efficiency. There is a need for analytical frameworks that can identify imbalance, anomalies, and risk zones within the Aadhaar ecosystem using publicly available UIDAI data.

Datasets Used

The analysis is based on the following UIDAI datasets:

Aadhaar Enrolment Dataset

Contains state- and district-level records of Aadhaar enrolments segmented by age groups.

Aadhaar Demographic Update Dataset

Includes updates related to address, date of birth, and other demographic attributes.

Aadhaar Biometric Update Dataset

Captures biometric update activity such as fingerprint and iris updates across regions.

Methodology Overview

The project follows a structured end-to-end analytical workflow:

- Data cleaning and preprocessing
- Harmonization of enrolment and update datasets
- Exploratory data analysis at state and district levels
- Construction of derived identity metrics
- Unsupervised machine learning for anomaly detection and clustering
- Policy-oriented interpretation of results

Machine learning is applied selectively to support insight generation and anomaly identification, not as a replacement for analytical reasoning.

Key Patterns Identified

The following patterns were consistently observed:

- Aadhaar enrolment volumes vary widely across districts, while update activity does not scale proportionally with enrolment.
- Several districts show high enrolment counts but very low demographic and biometric update activity, indicating potential identity maintenance gaps.

- Biometric updates are unevenly distributed, with certain regions showing disproportionately low biometric refresh activity.

- Update behavior differs significantly even within the same state, highlighting the importance of district-level analysis.

Identity Health Score (Core Contribution)

A composite **Identity Health Score** was developed to quantify the maintenance quality of Aadhaar identities at state and district levels.

Metric Definition

Identity Health Score=Demographic Updates + Biometric UpdatesEnrolments\text{Identity Health Score} = \frac{\text{Demographic Updates + Biometric Updates}}{\text{Enrolments}}Identity Health Score=EnrolmentsDemographic Updates + Biometric Updates

Interpretation

- **High score:** Active identity maintenance and operational stability

- **Low score:** Stagnant or outdated identity records

This score enables classification of regions into **Healthy**, **Moderate Risk**, and **High Risk** categories, supporting targeted administrative planning.

Enrolment–Update Imbalance Index (Unique Metric)

To further capture operational stress, an **Imbalance Index** was introduced:

Imbalance Index=EnrolmentsDemographic Updates + Biometric Updates + 1\text{Imbalance Index} = \frac{\text{Enrolments}}{\text{Demographic Updates + Biometric Updates + 1}}Imbalance Index=Demographic Updates + Biometric Updates + 1Enrolments

Insight

- **High imbalance:** Rapid enrolment without proportional maintenance

- **Low imbalance:** Stable and well-maintained identity ecosystem

This metric highlights districts where enrolment expansion may be outpacing update infrastructure.

Anomaly Detection Insights

Unsupervised machine learning (Isolation Forest) was used to identify anomalous regions.

Before Analysis

- Using enrolment counts alone, many regions appeared operationally normal.

After Analysis

- When Identity Health Score and update metrics were included, several districts and states emerged as anomalies.
- These anomalies reflect low update rates, unusual maintenance behavior, or structural imbalance rather than sheer volume.

This before-and-after comparison demonstrates how engineered metrics uncover hidden operational risks.

State-Level Behavioral Clustering

States were grouped using unsupervised clustering (KMeans) based on:

- Enrolments
- Demographic updates
- Biometric updates
- Identity Health Score

Cluster Outcomes

- **Mature Identity Ecosystems:** Balanced enrolment and update behavior
- **High Enrolment–Low Maintenance States:** Potential identity decay risk
- **Low Activity States:** Lower operational intensity

This segmentation supports differentiated policy and infrastructure strategies.

Quadrant Analysis

A quadrant-based visualization was used to classify states based on:

- X-axis: Enrolments
- Y-axis: Identity Health Score

This produces four intuitive operational categories:

- High enrolment / High health
- High enrolment / Low health
- Low enrolment / High health
- Low enrolment / Low health

This single visual effectively communicates ecosystem health to non-technical stakeholders.

Policy Recommendation Engine

A rule-based recommendation engine maps analytical insights to actionable interventions:

- **Low health score:** Awareness campaigns and update drives
- **High enrolment with low updates:** Mobile Aadhaar update units

- **Low biometric activity:** Biometric re-capture initiatives
- **Stable regions:** Maintain current operational strategy

This demonstrates how analytics can directly support governance decisions.

Limitations and Ethical Considerations

- The analysis uses aggregated, anonymized public data only.
- No individual-level or sensitive personal data is involved.
- The dataset represents a snapshot, limiting temporal trend analysis.
- Findings are intended for operational planning, not individual profiling.

Conclusion

This project demonstrates that Aadhaar functions as a living digital identity system requiring continuous maintenance rather than a one-time enrolment process. By introducing the Identity Health Score and Enrolment–Update Imbalance Index, and by applying unsupervised machine learning, the study provides a scalable framework for monitoring identity ecosystem health.

The approach respects data limitations while delivering actionable, policy-relevant insights, making it suitable for real-world governance and national infrastructure planning.

Code:
1. Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.ensemble import IsolationForest
```

2. Load UIDAI Datasets

```
enroll = pd.read_csv("aadhaar_enrolment.csv")
demo = pd.read_csv("aadhaar_demographic_updates.csv")
bio = pd.read_csv("aadhaar_biometric_updates.csv")
```

3. Basic Data Inspection

```
print(enroll.head())
print(enroll.info())

print(demo.head())
print(bio.head())
```

4. Aggregate Data at State–District Level

```
enroll_summary = enroll.groupby(
    ["state", "district"], as_index=False
)["enrolments"].sum()

demo_summary = demo.groupby(
    ["state", "district"], as_index=False
)["demo_updates"].sum()

bio_summary = bio.groupby(
    ["state", "district"], as_index=False
)["bio_updates"].sum()
```

5. Merge All Datasets

```
district_summary = enroll_summary.merge(
    demo_summary, on=["state", "district"], how="left"
).merge(
    bio_summary, on=["state", "district"], how="left"
)

district_summary.fillna(0, inplace=True)
district_summary.head()
```

6. Identity Health Score Calculation

```
district_data["identity_health_score"] = (
    district_data["demo_updates"] +
    district_data["bio_updates"]
```

```
) / (district_data["enrolments"] + 1)

district_summary[[
    "state", "district", "identity_health_score"
]].head()
```

	state	district	identity_health_score
0	Andhra Pradesh	Adilabad	629.000000
1	Andhra Pradesh	Alluri Sitharama Raju	411.500000
2	Andhra Pradesh	Anakapalli	515.000000
3	Andhra Pradesh	Anantapur	379.478261
4	Andhra Pradesh	Ananthapur	806.666667

7. Enrolment–Update Imbalance Index

```
district_summary["imbalance_index"] = (
    district_summary["enrolments"] /
    (district_summary["demo_updates"] +
     district_summary["bio_updates"] + 1)
)

district_summary[[
    "state", "district", "imbalance_index"
]].head()
```

8. Anomaly Detection (Isolation Forest)

```
features = district_summary[
    ["enrolments", "demo_updates", "bio_updates", "identity_health_score"]
]

scaler = StandardScaler()
scaled_features = scaler.fit_transform(features)

iso = IsolationForest(
    contamination=0.05,
    random_state=42
```

```
)

district_summary["anomaly"] = iso.fit_predict(scaled_features)

district_summary["anomaly"].value_counts()
```

```
anomaly
 1    765
-1     41
Name: count, dtype: int64
```

9. Identify Anomalous Districts

```
anomalies = district_summary[
    district_summary["anomaly"] == -1
]

anomalies[[
    "state", "district",
    "enrolments", "demo_updates",
    "bio_updates", "identity_health_score"
]].head()
```

10. State-Level Aggregation

```
state_summary = district_summary.groupby(
    "state", as_index=False
).agg({
    "enrolments": "sum",
    "demo_updates": "sum",
    "bio_updates": "sum",
    "identity_health_score": "mean"
})

state_summary.head()
```

11. Clustering States (KMeans)

```
cluster_features = state_summary[
    ["enrolments", "demo_updates", "bio_updates", "identity_health_score"]
]

scaled_cluster_features = scaler.fit_transform(cluster_features)
```

```
kmeans = KMeans(
    n_clusters=3,
    random_state=42
)

state_summary["cluster"] = kmeans.fit_predict(
    scaled_cluster_features
)

state_summary[["state", "cluster"]].head()
```

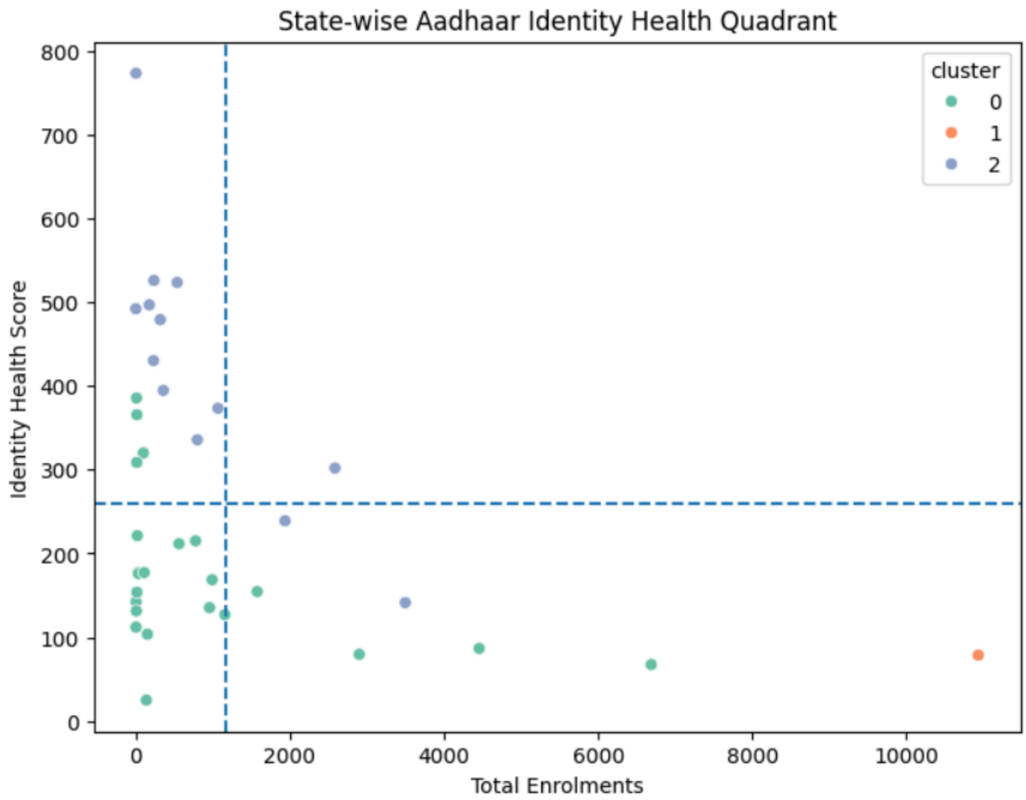
12. Quadrant Analysis Plot

```
plt.figure(figsize=(8,6))
sns.scatterplot(
    data=state_summary,
    x="enrolments",
    y="identity_health_score",
    hue="cluster",
    palette="Set2"
)

plt.axhline(
    state_summary["identity_health_score"].mean(),
    linestyle="--"
)

plt.axvline(
    state_summary["enrolments"].mean(),
    linestyle="--"
)

plt.title("State-wise Aadhaar Identity Health Quadrant")
plt.xlabel("Total Enrolments")
plt.ylabel("Identity Health Score")
plt.show()
```



13. Rule-Based Policy Recommendation Engine

```
def policy_recommendation(row):
    if row["identity_health_score"] < 0.05:
        return "Urgent update drive required"
    elif row["imbalance_index"] > 10:
        return "Deploy mobile update units"
    else:
        return "Operationally stable"

district_summary["policy_action"] = district_summary.apply(
    policy_recommendation, axis=1
)

district_summary[[
    "state", "district", "policy_action"
]].head()
```